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Does public data availability matter for entrepreneurship? Evidence from China's government data platform access

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Data are the "oil" of the digital economy. Using 278 prefecture-level cities in China, this study examines the impact of public data availability on regional entrepreneurship based on a quasi-natural experiment of open government data (OGD). The study finds that OGD exerts a positive effect on entrepreneurship. The mechanism analysis reveals that OGD encourages entrepreneurial activities by improving the entrepreneurial elements, including the entrepreneurial intention, opportunity, and capital. Further analysis reveals that OGD significantly impacts opportunity-driven entrepreneurship, the service sector, and low administrative level cities. Moreover, there is a notable policy synergy between OGD and the innovative city pilot policy, which together stimulate increased entrepreneurial activities. This study provides empirical support for sharing government data and offers important insights into fostering market vitality in the digital era.

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Introduction

In the era of the digital economy, the integration of information technology and the real economy has triggered the rapid growth of data. According to International Data Corporation forecasts, the world will generate 384.6 zettabytes of data by 2028, with a significant portion controlled by government agencies (Magalhaes and Roseira 2020). Since the U.S. established the open data website (www.data.gov) in 2009, the international community has actively engaged in the open government data (OGD) movement (Bates 2014). OGD involves making accessible to the public data resources generated, collected, stored, and processed by the government during the discharge of its duties. This initiative aims to encourage society's full utilization of these data to realize the economic value of the data elements. With the second largest data output, China started constructing a government data open platform in 2012 and has established such platforms in 204 cities as of August 2023. As a representative of developing countries, China has seen a phase of economic slowdown and a move toward high-quality economic development in recent years, facing challenges of economic transition and middle-income trap. According to some scholars, fostering an entrepreneurial spirit and promoting entrepreneurship are essential sources of motivation to stimulate economic growth and overcome the middle-income trap (Baumol and Strom 2007; Wright and Stigliani 2013). Thus, this study explores the potential for OGD to help developing countries overcome the middle-income trap from the perspective of regional entrepreneurship.

Most studies prove that entrepreneurship is influenced by formal institutions (Minniti 2008; Obaji and Olugu 2014; Kong and Qin 2021; Díez-Martín et al. 2022; Prasannath et al. 2024; Wang et al. 2025), informal institutions (Harms and Groen 2017; Xiong et al. 2023), infrastructure construction (Li et al. 2023; Li et al. 2024; Zhang et al. 2024), and climate change (Han and Zhou 2025). It can be observed that evidence to prove whether the OGD movement can influence entrepreneurs' decision-making is still lacking. However, given the significant role of entrepreneurship in economic growth and overcoming the middle-income trap (Baumol and Strom 2007; Wright and Stigliani 2013; Singh and Ashraf 2020), we attempt to empirically examine their relationship from the perspective of Chinese cities, aiming to provide valuable insights for the economic transition of more developing countries like China.

We chose to study this issue in China for two main reasons. First, China offers an ideal setting to explore the causal relation between OGD and entrepreneurship. The introduction of an online open data platform by local governments as an exogenous shock to potential entrepreneurs allows us to identify its impact on entrepreneurial activities. Second, there are significant regional differences in entrepreneurial activities across cities. To address this, we employ the quasi-natural experiment and match city-level variables with business registration data to examine how OGD affects regional entrepreneurship. This study focuses on the following: (1) Does OGD stimulate more entrepreneurial activities? (2) How does OGD influence entrepreneurial activity? (3) Is there heterogeneity in the impact of OGD on regional entrepreneurial activity? Specifically, under what conditions does OGD more effectively promote entrepreneurship? (4) In the context of innovation-driven entrepreneurship, is there a synergy between OGD and innovation policies?

This study makes the following contributions to the literature. First, it expands the value effects of OGD from the perspective of regional entrepreneurial activities. Prior studies paid more attention to the social value of OGD, such as information justice (Johnson 2014), accountability and transparency (Lourenço 2015; Saxena and Muhammad 2018), and citizen participation (Wirtz et al. 2019). We reveal that OGD can encourage new firm entry to generate economic value. Second, fostering entrepreneurship is crucial for cultivating new drivers of economic growth and overcoming the middle-income trap. This study uses a staggered difference-in-differences (DID) model to discuss how big data influences the entrepreneurial environment and provides insights into how to enhance entrepreneurial activities in the digital economy. Third, as a typical developing country, using the Chinese OGD initiatives to stimulate entrepreneurial activities offers a new development model that may help emerging economies to effectively realize the potential of public data and provide valuable insights for overcoming the middle-income trap.

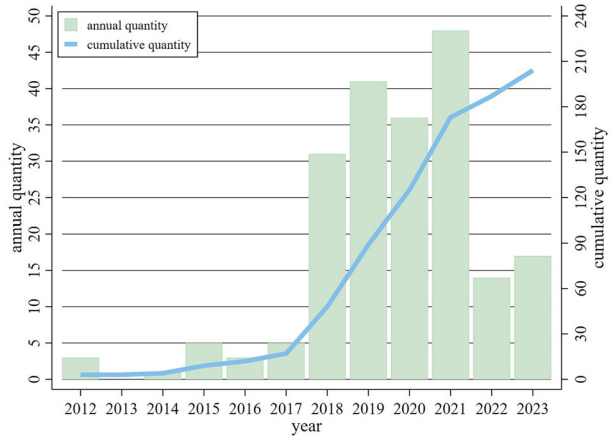


Fig. 1 Number of OGD platforms in China, 2012–2023.

Background and research hypothesis

Background. Public data owned by the government covers many fields, including economic construction, public security, resources and the environment, traffic, medical care, credit services, taxation and finance, health, culture, and leisure. The Chinese government attaches great importance to the accessibility and application of public data, Shanghai launched China's first government public data open platform in 2012, and Zhejiang Province successfully launched China's first provincial government public data open platform in 2015. The State Council issued the "Action Plan for Promoting Big Data Development" in 2015, proposing to promote the opening of public data and accelerate the speed of building an OGD platform. Subsequently, local governments have actively carried out the construction and launch of online OGD platforms. In 2022, the State Council issued the "Guiding Opinions on Strengthening the Construction of Digital Government," making "building an open and shared data resource system" one of the key efforts. Under policy guidance, as of August 2023, 204 local governments (61.26%) of China's cities had launched open data platforms. The number and distribution for OGD platforms across prefecture-level cities is shown in Figs. 1 and 2.

Research hypothesis. Entrepreneurship involves potential entrepreneurs discovering, identifying, and pursuing untapped business opportunities—a process inevitably shaped by the external entrepreneurial environment (Zahra et al. 2014). Thus, a supportive entrepreneurial environment is crucial for enhancing the potential and willingness of market participants (Edelman and Yli–Renko 2010; Alvarez et al. 2011). OGD has a significant impact on this environment, such as improving government transparency (Saxena and Muhammad 2018; Cahlikova and Mabillard 2020), reducing the perception of uncertainty (Xu et al.

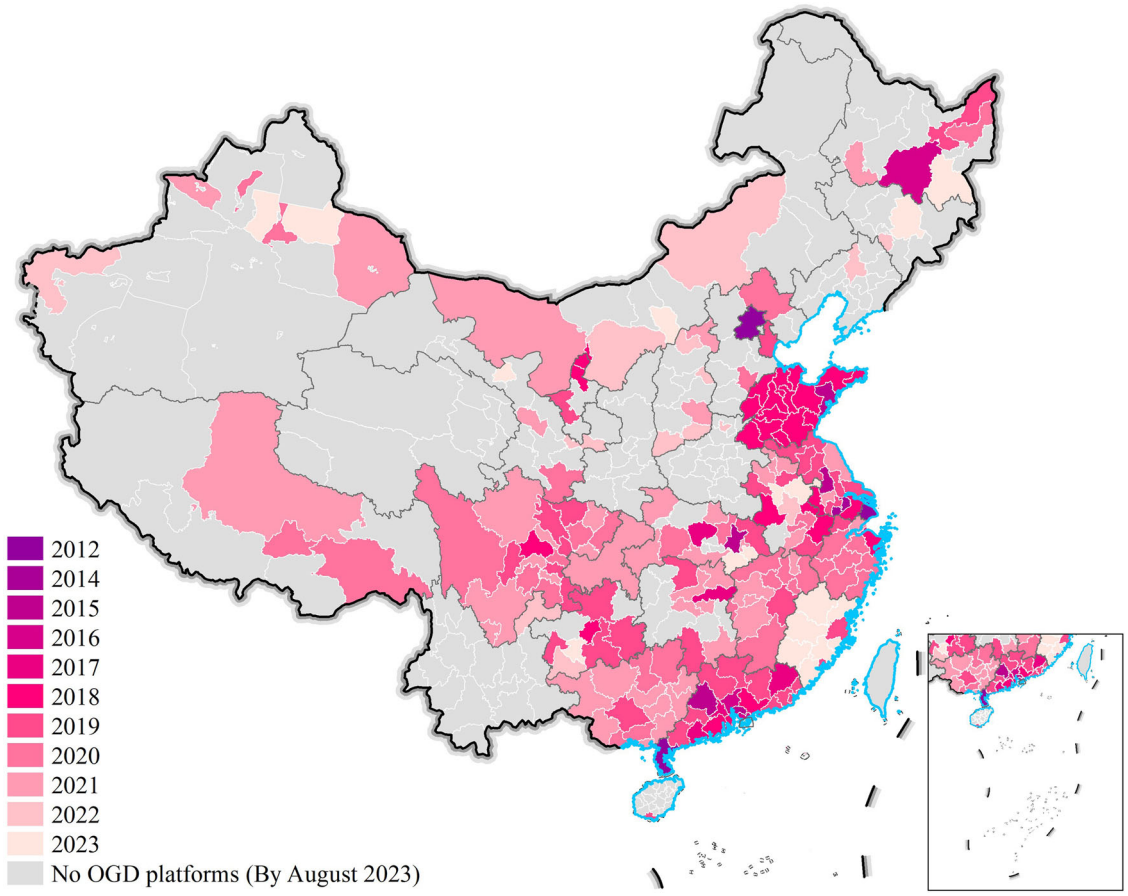


Fig. 2 Distribution of OGD platforms in China, 2012–2023.

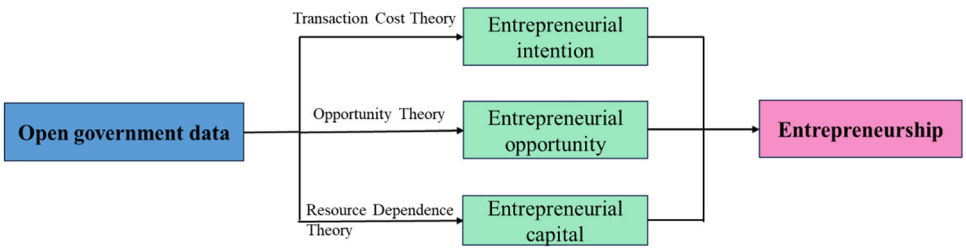


Fig. 3 Theoretical Framework.

2024), and lowering finance costs (Xing et al. 2024). This study examines how OGD affect entrepreneurship, focusing on the three factors necessary to start a business: entrepreneurial intention, opportunity, and capital. The research framework is shown in Fig. 3.

First, access to public sector information through online open data platforms allows society to better understand and monitor government's decision-making process, thereby enhancing government transparency and efficiency (Lourenço 2015; Kochanova et al. 2020; Park and Gil-Garcia 2022). OGD lowers the institutional transaction costs of new firms' entry, mainly manifested in reducing rent-seeking behavior and decreasing nonproductive expenditures. According to transaction cost theory, publicly available government data are conducive to creating a suitable institutional environment for entrepreneurship and stimulating entrepreneurial intentions, ultimately encouraging entrepreneurial activities (Bowen, De Clercq (2008); Bosma and Schutjens 2009; Urbano and Alvarez 2014; Wang and Zhang 2025).

Second, opportunity theory conceptualizes entrepreneurship as a comprehensive process involving the effective identification of potential entrepreneurial opportunities and their subsequent development and exploitation (Shane and Venkataraman 2000). OGD offers potential entrepreneurs low-cost information sources, helping entrepreneurs better understand local social and business conditions, government policies, and market demands, such as determining a company's location, understanding consumer preferences, and innovating business models (Nagaraj 2022; Huber et al. 2022; Bilan et al. 2023). Thus, entrepreneurs can search, download, and analyze public data from online platforms, which will greatly expand their cognition range of opportunity exploitation and decision-making (Hughes-Cromwick and Coronado 2019; Jetzek et al. 2019; Saeedikiya et al. 2024). A typical case is when Alex Laskey and Daniel Yates used U.S. government-released energy consumption data to create Opower—a company that provides energy planning services to help households reduce waste (Jetzek et al. 2014). The availability of public data can reduce the perception of uncertainty caused by

information asymmetry and identify market opportunities accurately, which will eventually promote entrepreneurship (Corbett 2007; Hegde and Tumlinson 2021; Liu et al. 2025). Moreover, OGD is mutually reinforcing with the development of new infrastructures such as 5G networks and arithmetic centers, thereby improving a city's digital infrastructure with the progress of OGD and attracting more entrepreneurs (Schade and Schuhmacher 2022; Li et al. 2023; Salamzadeh et al. 2024; Zhu et al. 2025).

Finally, resource dependence theory emphasizes an organization's dependence on external resources, asserting that its survival and development are contingent upon its ability to acquire and manage these resources, while OGD affects enterprises' access to resources. We consider the entrepreneurial capital, focusing on the venture capital and trade credit. In the context of information asymmetry, venture capitalists tend to invest in cities with higher social credit to mitigate uncertainty and reduce investment risks. OGD, including lists of dishonesty and administrative penalties, can effectively mitigate opportunistic behavior among transaction partners and build a high-quality urban social credit system (Fan et al. 2023). Under the influence of reputation mechanisms, the availability of credit data strengthens the awareness of integrity among different economic entities and creates a favorable commercial credit environment (Liang et al. 2018). This will attract more venture capital and improve the external financing capabilities of new start-ups (Schmidthuber et al. 2021; Xing et al. 2024), significantly enhancing enterprises' resource acquisition capacity and driving entrepreneurial activities (Qin and Kong 2022; Fang et al. 2023).

Therefore, this study proposes the following research hypothesis:

Hypothesis 1. OGD can significantly promote the regional entrepreneurship.

Research design

Data. This study uses data on 278 prefecture-level cities from 2009 to 2023, excluding non-city administrative units (such as regions, autonomous prefectures, and leagues) and those that experienced administrative boundary changes during the research period. The final sample comprised 4,104 observations. Newly registered firms' data were sourced from the Tianyancha platform, with other data obtained from the "China City Statistical Yearbook", the Economy Prediction System (EPS) database, and the China Stock Market and Accounting Research (CSMAR) database. All continuous variables were winsorized at the 1% and 99% levels.

Independent variable. Open Government Data (*OGD*). This study adopts the method used by Xing et al. (2024). Based on the "China Local Government Data Open Report (Urban)" provided by Fudan University's Digital and Mobile Governance Laboratory, we manually verified the data using search engines, including Baidu. If a city launched a public data open platform in a given year, the *OGD* value for that year and subsequent years is set to 1 and 0 otherwise.

Dependent variable. Entrepreneurship (*Entrep*). Entrepreneurial activity is a key indicator of a city's entrepreneurship. Considering measurement bias arising from heterogeneity in city size, we use the city's registered population as the standardization base and measure regional entrepreneurial vitality by the number of newly registered firms per 100 people (Gan et al. 2023).

Control variables. Drawing on existing literature (Kong and Qin 2021; Gan et al. 2023), we selected several control variables, including economic development level (*Pgdp*), traditional financial resources (*Finan*), human capital (*Human*), foreign investment level (*Foreign*), digital financial inclusion (*Digfin*), intellectual property protection (*Ipp*), marketization degree (*Market*), and fiscal decentralization (*Fiscal*).

Moreover, considering that the launch of government data open platforms in various cities may not be random, local governments may need to make decisions based on regional economic development levels, geographical locations, and resource endowments. These factors, which evolve over time, could influence regional entrepreneurial vitality, potentially rendering the identification assumption of the multi-period DID model invalid. Therefore, we further incorporate the per capita grain production in 2000, per capita oilseed output in 2000, per capita vegetable production in 2000, the number of regular higher education institutions in 2000, and time trend interaction terms to control for pre-existing socio-economic characteristics that influence whether a city establishes a government data open platform. This approach thereby mitigates estimation bias arising from the selection of the treatment group.

Model specification. This study utilizes the staggered launch of government data open platform as an exogenous shock to construct Model (1) and analyze the impact of open government data on regional entrepreneurship.

$$Entrep_{i,t} = \partial_0 + \partial_1 OGD_{i,t} + \sum_k \partial_k X_{k,i,t} + \mu_i + \theta_t + \varepsilon_{i,t} \quad (1)$$

In the model, *Entrep* represent entrepreneurship. *OGD* indicates whether a city launched a government data open platform current year. μ_i denotes city fixed effects, θ_t denotes year fixed effects, and $\varepsilon_{i,t}$ represents the random error term. The standard errors are clustered at the city level. Definitions of the variables are provided in Table 1.

Empirical results

Descriptive statistics and univariate analysis. Table 2 provides the descriptive statistics of the variables. The mean for *Entrep* is 1.279, with a minimum of 0.147 and a maximum of 5.302. These figures reflect a strong entrepreneurial spirit in China, although there is considerable variation in activities across cities. The mean value for *OGD* is 0.194, indicating that 19.4% of the cities in the sample are in the treatment group. The median value is 0, indicating that OGD is still emerging in China.

As shown in Fig. 4, we can find that there is no severe multicollinearity among the variables. And we conduct the variance inflation factor (VIF) test, the mean VIF is 2.13, lower than the critical value of 10.

Baseline results. Table 3 reports the estimated results based on Model (1). The analysis reveals a significant positive correlation between *OGD* and *Entrep*, even when controlling for other factors. Cities with OGD have a higher number of new enterprises, indicating that OGD can stimulate more entrepreneurship, thereby confirming Hypothesis 1.

Parallel trends test. The parallel trends assumption is crucial for a DID analysis. Figure 5 depicts the results of the parallel trends test with 95% confidence intervals. It can be seen that the regression coefficients before the OGD all fluctuate around zero and are statistically insignificant. This indicates that no significant pre-existing trend differences in regional entrepreneurial vitality existed between the treatment and control groups prior to the OGD, thus consistent with the parallel trend assumption. The policy effect manifests with a one-period lag

Table 1 Definition and descriptions of variables.

Type	Variable	Definition
Dependent variable	*Entrep*	The number of newly registered firms per 100 people in a city
Independent variable	*OGD*	If a city launched a government data platform that year, the value is 1; otherwise, it is 0.
	Pgdp	The natural logarithm of per capita GDP in the city
	Finan	The ratio of the loan balance of financial institutions to GDP in the city
	Human	The ratio of the number of regular undergraduate students in the city to the resident population
	Foreign	The ratio of total local import and export volume to GDP
	Digfin	The digital financial inclusion index from Peking University divided by 100
Control variables	*Ipp*	(Number of intellectual property cases closed in the city/City GDP)/(Total number of intellectual property cases closed in China/National GDP)
	Market	The ratio of government general fiscal expenditures to GDP
	Fiscal	The ratio of government general fiscal revenue to general fiscal expenditures
	Pcgp	The ratio of total grain production to registered population
	Pcop	The ratio of total oilseed output to registered population
	Pcvp	The ratio of total vegetable production to registered population
	Educ	The number of regular higher education institutions in the city

Table 2 Descriptive statistical results of variables.

Variable	N	Mean	Min	Median	Max	Sd
Entrep	4104	1.279	0.147	1.013	5.302	0.952
OGD	4104	0.194	0.000	0.000	1.000	0.396
Pgdp	4104	10.736	8.842	10.731	12.087	0.614
Finan	4104	1.488	0.544	1.361	3.621	0.617
Human	4104	0.020	0.000	0.011	0.127	0.026
Foreign	4104	0.183	0.002	0.081	1.827	0.282
Digfin	4104	1.430	0.082	1.472	2.746	0.789
Ipp	4104	0.171	0.000	0.073	1.413	0.263
Market	4104	0.193	0.074	0.170	0.620	0.092
Fiscal	4104	0.453	0.089	0.421	0.998	0.218
Pcgp	4104	0.383	0.015	0.377	0.916	0.166
Pcop	4104	0.023	0.000	0.016	0.107	0.022
Pcvp	4104	0.361	0.008	0.290	1.191	0.232
Educ	4104	3.865	0.000	2.000	32.000	6.062

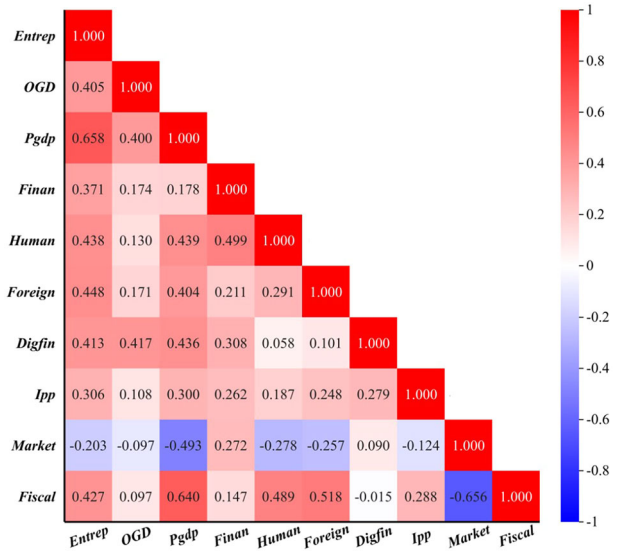


Fig. 4 Correlation coefficients of the main variables.

after implementation, after which it loses statistical significance. Consequently, future efforts should place greater emphasis on sustaining the effects of OGD on regional entrepreneurial vitality.

Heterogeneity treatment effect testing. The heterogeneity of treatment effects in the multi-period DID model with two-way fixed effects estimation may lead to bias due to effect heterogeneity, where the same model produces differential effects across individuals. To address this, we conduct a test using a heterogeneity-robust estimator. First, Goodman-Bacon (2021) argues that control groups consist of three types: "never-treated individuals," "not-yet-treated individuals," and "already-treated individuals". Among these, the third type constitutes a bad control group. If the proportion of such individuals is too high, the estimation results may be significantly biased. Therefore, we conducted a Bacon decomposition using balanced panel data from 2009 to 2023. Table 4 shows that the weight of the "never-treated individuals" as the control group in the time-varying treatment group is approximately 55%. Figure 6 shows that the "bad treatment group" (using the treated group as the control) that may lead to overestimation accounts for only 5.54%, indicating that treatment effects exhibit a certain degree of heterogeneity across groups but do not cause severe bias in the estimation results.

Second, we followed the approach of De Chaisemartin, d'Haultfoeuille (2020) for the decomposition. Table 5 reports the decomposition results, showing that the total weight is 798, including 770 positive weights and 28 negative weights, with negative weights accounting for only 3.51%. This indicates that the estimates do not exhibit severe bias issues.

Third, the first two methods only diagnosed the treatment effect of heterogeneity but did not address this bias. Therefore, based on the "stacked DID" proposed by Cengiz et al. (2019), we restructured the dataset into balanced panel data relative to the event time. By controlling for group effects and time fixed effects, we obtained the weighted average of the treatment effect, which can address the two-way fixed effect estimation bias. Figure 7 shows that there was no significant trend in differences in regional entrepreneurial vitality between the treatment and control groups prior to the opening of government data.

Endogeneity test

Placebo test. To determine if the observed increase in regional entrepreneurial activities is truly due to the OGD rather than other policies or random factors, this study uses a random sampling method to control for random effects. Specifically, a subset of cities is randomly selected as a pseudo-treatment group, while the remaining cities act as a pseudo-control group. Model (1) is run 500 times on this setup. Figure 8 depicts the distribution of the regression coefficients. These coefficients, from the random samples, are normally distributed around 0 and differ significantly

Table 3 Baseline regression results.

Variable	(1) *Entrep*	(2) *Entrep*	(3) *Entrep*
OGD	0.195***	0.129**	0.110**
	(3.26)	(2.24)	(2.00)
Pgdp		−0.173	0.013
		(−1.31)	(0.11)
Finan		−0.389***	−0.341***
		(−3.88)	(−3.83)
Human		4.747	5.008
		(1.09)	(1.36)
Foreign		0.026	0.152
		(0.11)	(0.63)
Digfin		0.087*	0.035
		(1.77)	(0.78)
Ipp		0.121**	0.086
		(2.03)	(1.57)
Market		−0.185	0.258
		(−0.47)	(0.66)
Fiscal		−0.089	0.005
		(−0.44)	(0.03)
c.Pcgp#c.year			−0.041
			(−1.56)
c.Pcop#c.year			−0.085
			(−0.67)
c.Pcvp#c.year			0.032**
			(2.38)
c.Educ#c.year			0.003***
			(3.54)
Constant	1.241***	3.518**	−8.176
	(106.83)	(2.33)	(−0.39)
City FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	4104	4104	4104
Adj R^2	0.743	0.753	0.761

The values in parentheses are t-values adjusted for clustering at the city level. "*", "**" and "***" denote significance at the 10%, 5%, and 1% levels, respectively.

from the baseline coefficients (0.110) reported earlier. This implies that the baseline results are indeed due to the OGD.

Instrumental variable test. The launch of government data open platforms may be related to local economic, social, and cultural factors, which also influence regional entrepreneurial vitality. Therefore, to mitigate endogeneity bias caused by omitted variables, we selected the interaction term between terrain undulation and the logarithm of broadband Internet access users in the prior period as an instrumental variable (*IV*) for two-stage least squares testing. This is mainly because the flatness of the regional terrain directly affects the construction of local network infrastructure, and the number of broadband Internet access users reflects the level of local Internet development, both of which affect the launch of government data open platforms. However, terrain undulation and the number of broadband Internet users in the previous period are unlikely to directly affect the entrepreneurial behavior of market entities in the current period.

As shown in columns (1) and (2) of Table 6, the coefficient of *IV* in the first stage is negative, indicating that the steepness of the regional terrain has a negative impact on the construction of government data open platforms. The coefficient of *OGD* in the second stage is positive, which is in line with expectations, and the Cragg-Donald-Wald F value is 111.08, which is greater than the empirical value of 10, ruling out the possibility of weak instrumental variables.

PSM-DID. The multiple-period DID method can identify the average treatment effect of policy implementation, but the availability of government data may lead to endogeneity problems caused by selection bias. We further use a propensity score matching model to mitigate the problem of sample selection bias. Specifically, we selected the control variables from Model (1) as covariates for a 1:3 nearest neighbor matching with a caliper of 0.05. The matched samples were regressed using Model (1). The result in column (3) of Table 6 shows that the regression coefficient for *OGD* remains significantly positive, indicating the robustness of the baseline regression results.

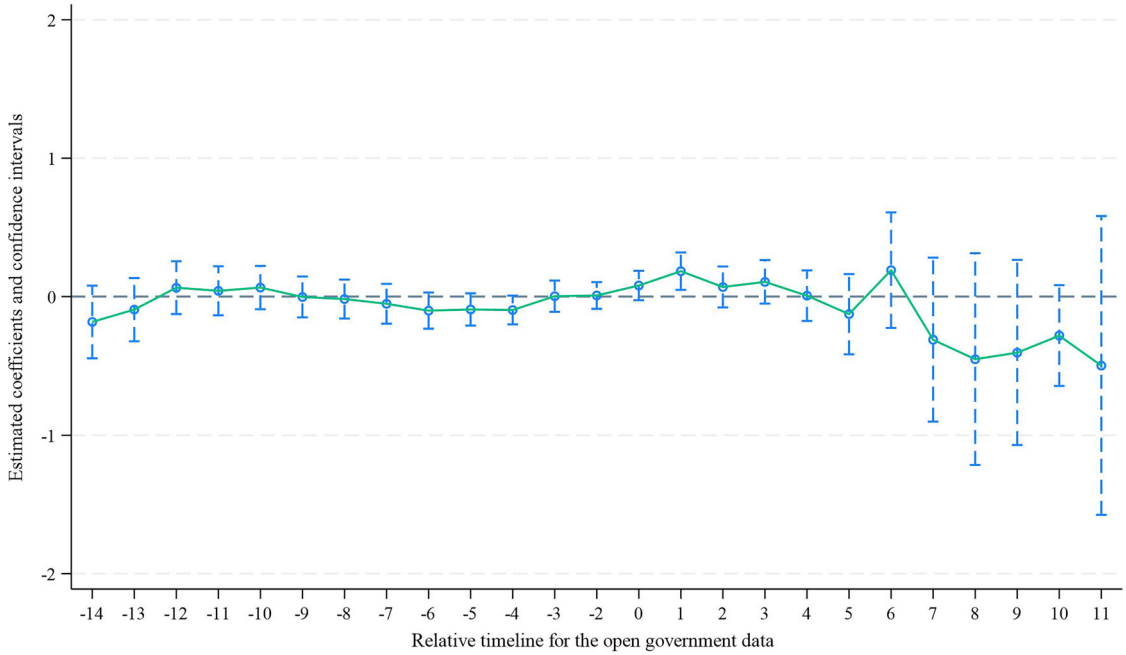


Fig. 5 Parallel trend test.

DML causal test. Compared with traditional causal inference methods, the advantages of the dual machine learning method lie in variable selection and model specification. Following the approach of Bodory et al. (2022), we set the sample splitting ratio for the dual machine learning method to 1:4, employs a general interactive dual machine learning model for testing, and uses the random forest algorithm for prediction. Under the control of city and year fixed effects, the model progressively incorporates the linear, quadratic, and cubic terms of the control variable set. The results in columns (4) to (6) of Table 6 show that the regression coefficients of *OGD* are significantly positive at the 1% level. The above results indicate that after using a dual machine learning method to mitigate potential endogeneity issues, the research conclusions remain robust.

Control for confounding policy effects. During the sample period, several policy changes such as the OGD can affect the research findings. We manually collected data on related events, including the Broadband China pilot, Smart City pilot, and National Information Benefits People pilot, to control for confounding policy effects. First, we create policy dummy variables as follows: If a city was designated as a Broadband China pilot, Smart City pilot, or National Information Benefits People pilot in year *t*, then *Band*, *Smart*, or *Info* equals 1 for that year and subsequent years and 0 otherwise. We then added each of these dummy variables (*Band, Smart*, and *Info*) individually and together in Model (1). Table 7 presents the results, indicating that the coefficient of *OGD* remains significantly positive. This implies that the impact of OGD on entrepreneurship is not influenced by other similar policies during the same period.

Other robustness tests. To ensure the robustness of the research findings, we conducted several additional checks: First, we employed an alternative measurement of explanatory variables. As the rollout of OGD platforms at the provincial level (including municipalities) will impact cities in the affected provinces, we redefined *OGD* as follows: if a city's province has launched a public data open platform, *OGD* equals 1 for that year and subsequent years and 0 otherwise. We also controlled for provincial fixed effects and adjusted standard errors at the provincial level. The result is presented in column (1) of Table 8. Second, to control for the impact of the inherent characteristics of cities on their entrepreneurial vitality, we added a city-year trend term to Model (1). The result is shown in Table 8, column (2). Third, under China's vertically structured administrative management system, provincial governments often exert significant influence on the decision-making of cities within their jurisdictions. We incorporated province-year interaction fixed effects into Model (1) to control for this factor, with the result shown in Table 8, column (3). The coefficients of *OGD* in the above results are significantly positive, implying that the results are robust.

Analysis of mechanisms. Building on the theoretical analysis, the following section explores how OGD impacts entrepreneurship across the following three key dimensions for starting a new business.

Entrepreneurial intention. According to the transaction cost theory, high institutional transaction costs can dampen potential entrepreneurs' expectations and willingness, thereby hindering local entrepreneurial activities. (Lu and Tao 2010; Audretsch et al. 2022). We measure entrepreneurial intention in two ways, one is the ratio of the sum of management, financial, and sales expenses to the total assets for all listed companies in the same prefecture-level city for a given year. The other is the Baidu search index, using the natural logarithm of the following keywords related to the prefecture-level cities: innovation, entrepreneurship, startup, startup company, venture capital, business plan, business model, business risk, financing, market, profit, and team. As presented in columns (1) and (2) of Table 9, the coefficient of *OGD* with

Table 4 Bacon decomposition results.

Bacon Decomposition	Beta	Total weight
Timing_groups	0.179	0.392
Never_vs_timing	0.181	0.552
Within	−1.268	0.055

Table 5 De Chaisemartin and D' Haultfoeuille decomposition results.

Total	Positive weights	Negative weights	Positive weights ratio	Negative weights ratio
798	770	28	96.49%	3.51%

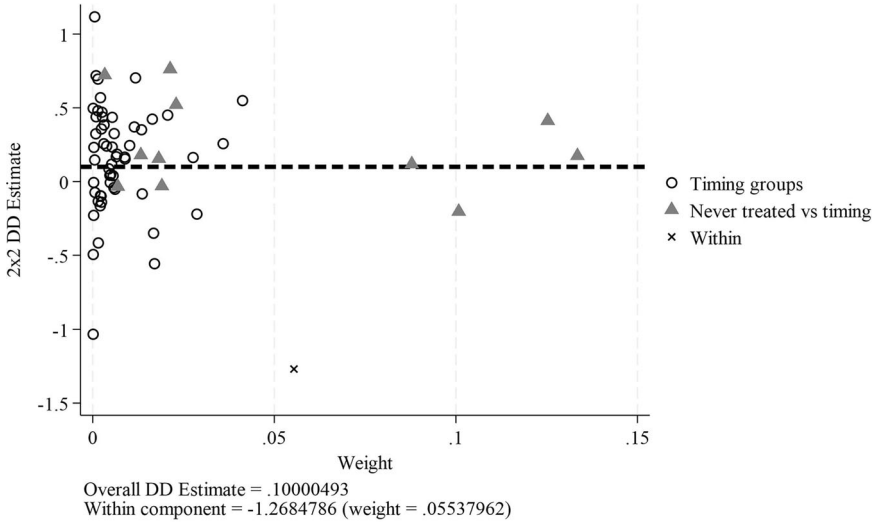


Fig. 6 Goodman-Bacon decomposition.

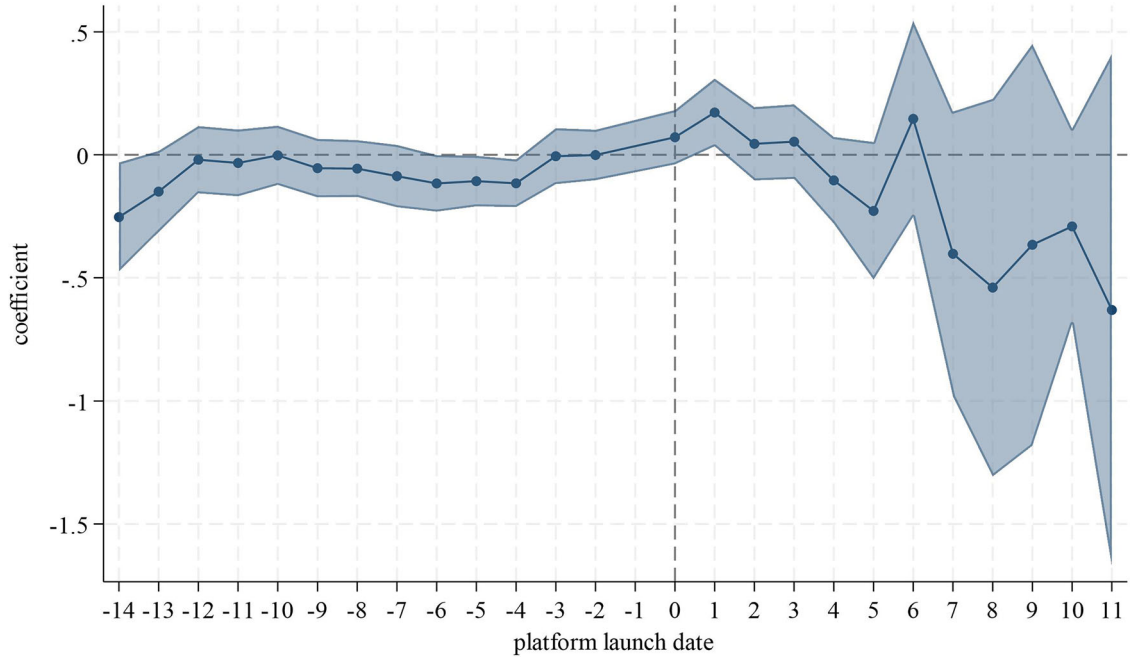


Fig. 7 Event study (Cengiz et al. 2019).

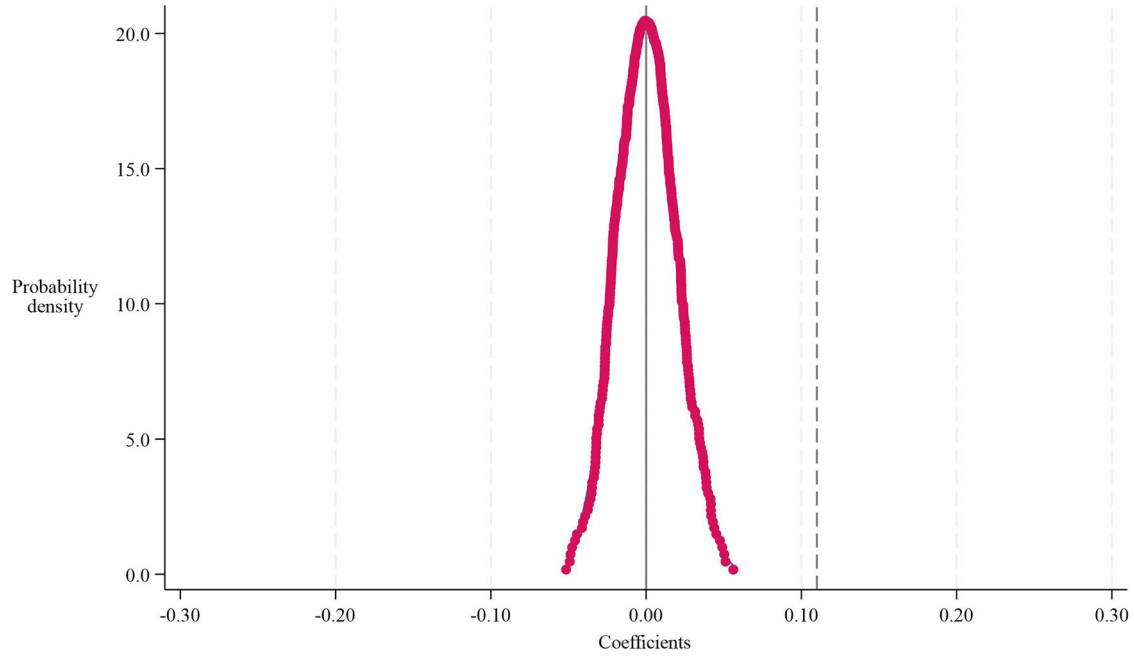


Fig. 8 Placebo test.

Institutional cost is significantly negative, while with *Baidu index* is significantly positive, implying that OGD effectively reduce regional institutional transaction costs and stimulate entrepreneurial enthusiasm.

Entrepreneurial opportunity. Information asymmetry often limits entrepreneurs' understanding and reduces their entrepreneurial motivation (Corbett 2007; Hegde and Tumlinson 2021). As public information availability is difficult to measure directly, we used the patent applications instead. It has been documented that information availability promotes innovation, which is the engine of entrepreneurship (Brown and Martinsson 2019; González-Pernía et al. 2015). Well-developed infrastructure boosts connectivity and information flow, which supports entrepreneurial activity (Schade and Schuhmacher 2022; Li et al. 2024), and we assessed the digital infrastructure using the number of internet access points per 100 people. As presented in columns (3) and (4) of Table 9, the regression coefficients of *OGD* are significantly positive, indicating that OGD effectively reduce information asymmetry and improve local digital infrastructure, help entrepreneurs better innovate and identify market opportunities.

Entrepreneurial capital. A favorable financing condition means that start-ups can easily access funding and better meet entrepreneurs' capital needs, ultimately stimulating public enthusiasm

Table 6 Endogeneity test results.

Variable	(1) *OGD*	(2) *Entrep*	(3) *Entrep*	(4) *Entrep*	(5) *Entrep*	(6) *Entrep*
OGD		0.939**	0.087*	0.356***	0.334***	0.351***
		(2.48)	(1.85)	(21.27)	(10.87)	(18.85)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
IV	−0.074***					
	(−4.71)					
Constant	−1.383		−11.807			
	(−0.11)		(−0.55)			
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls quadratic terms	No	No	No	No	Yes	Yes
Controls cubic terms	No	No	No	No	No	Yes
N	4088	4088	3964	4104	4104	4104
Adj R^2	0.585		0.720			

The values in parentheses of columns (1)-(3) are t-values adjusted for clustering at the city level. The values in parentheses of columns (4)-(6) are robust z-values."*", "**" and "***" denote significance at the 10%, 5%, and 1% levels, respectively.

Table 7 Regression results excluding confounding policy interference.

Variable	(1) *Entrep*	(2) *Entrep*	(3) *Entrep*	(4) *Entrep*
OGD	0.109*	0.110**	0.103*	0.103*
	(1.96)	(2.00)	(1.86)	(1.86)
Band	0.021			−0.008
	(0.36)			(−0.14)
Smart		0.007		−0.007
		(0.14)		(−0.15)
Info			0.169**	0.172**
			(2.35)	(2.34)
Controls	Yes	Yes	Yes	Yes
Constant	−8.215	−8.100	−6.755	−6.793
	(−0.39)	(−0.39)	(−0.32)	(−0.32)
City FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	4104	4104	4104	4104
Adj R^2	0.761	0.761	0.762	0.762

The values in parentheses are t-values adjusted for clustering at the city level. "*" and "**" denote significance at the 10% and 5% levels, respectively.

Table 8 Other robustness regression results.

Variable	(1) *Entrep*	(2) *Entrep*	(3) *Entrep*
OGD	0.119*	0.116**	0.053*
	(1.95)	(2.09)	(1.84)
Controls	Yes	Yes	Yes
Constant	−9.991	38.271	−47.728***
	(−0.36)	(1.32)	(−3.29)
City FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Province FE	Yes	No	No
City-Year Trend	No	Yes	No
Province-Year FE	No	No	Yes
N	4104	4104	4018
Adj R^2	0.759	0.762	0.840

The values in parentheses are t-values adjusted for clustering at the city level. "*", "**" and *** denote significance at the 10%, 5%, and 1% levels, respectively.

for entrepreneurship. We use the amount of venture capital per 100 people in a city to measure the attractiveness of city to draw venture capital and measure urban trade credit using the China Urban Commercial Credit Environment Index. As presented in columns (5) and (6) of Table 9, the coefficients of *OGD* are significantly positive, indicating that OGD can substantially improve urban credit conditions and support financing for entrepreneurs.

Further analysis

Heterogeneity analysis

Heterogeneity of entrepreneurial motivations. In 2001, the Global Entrepreneurship Monitor categorized entrepreneurial activities into necessity- and opportunity-driven entrepreneurship based on motivation. Necessity-driven entrepreneurship arises when individuals start businesses to meet basic needs, often in low-barrier industries such as food and retail, leading primarily to self-employment (Block and Sandner 2009). On the contrary, opportunity-driven entrepreneurship involves actively seeking or creating new business opportunities to achieve personal goals, usually resulting in greater job creation. Opportunity-driven entrepreneurs typically face higher information barriers and greater risks of failure and are more sensitive to the quality of the regional entrepreneurial environment (Boudreaux and Nikolaev 2019; Huang et al. 2023). Conversely, necessity-driven entrepreneurs face lower market entry barriers and have less demand for information resources (Millán et al. 2012). Therefore, we argue that OGD are more likely to stimulate opportunity-driven entrepreneurship.

To verify this, we classified newly registered enterprises into necessity- and opportunity-driven entrepreneurship. Then, we measured the former with individual enterprises and the latter with nonindividual enterprises. Using entrepreneurship type as the classification criterion, we conducted regression analysis with Model (1). The results are presented in columns (1) and (2) of Table 10, revealing that *OGD* are only positive and significantly correlated with *Entrep_Oppo*. This implies that OGD have a more pronounced effect on fostering opportunity entrepreneurship compared with necessity entrepreneurship.

Heterogeneity of industry types. Due to differing foundational conditions and sensitivities to the entrepreneurial environment across industries, the impact of OGD on entrepreneurship may vary by industry. Generally, the manufacturing sector has higher

Table 9 Mechanism analysis regression results.

	(1)	(2)	(3)	(4)	(5)	(6)
	Entrepreneurial intention		Entrepreneurial opportunity		Entrepreneurial capital	
Variable	*Institutional cost*	*Baidu index*	*Patent application*	*Digital infrastructure*	*Venture capital*	*Trade credit*
OGD	−0.007**	0.024***	0.094***	0.005**	0.038***	0.927***
	(−2.33)	(3.98)	(3.65)	(2.12)	(2.87)	(4.38)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Constant	1.990	0.015	8.096	−6.437**	−1.035	19.352
	(1.40)	(0.01)	(1.11)	(−2.43)	(−0.38)	(0.19)
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	3423	4104	4104	4045	4104	4097
Adj R^2	0.687	0.943	0.838	0.796	0.553	0.739

The values in parentheses are t-values adjusted for clustering at the city level. "*", "**" and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 10 Heterogeneity analysis results.

	(1)	(2)	(3)	(4)	(5)	(6)
Variable	*Entrep_Nece*	*Entrep_Oppo*	*Entrep_Manu*	*Entrep_Serv*	*Entrep_High*	*Entrep_Low*
OGD	0.070	0.114***	0.005	0.149*	−0.067	0.123**
	(1.20)	(3.40)	(1.18)	(1.83)	(−0.48)	(2.15)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Constant	−18.493	8.673	−1.548	−0.928	−15.553	−2.117
	(−1.17)	(0.82)	(−1.15)	(−0.04)	(−0.20)	(−0.09)
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	4104	4104	4104	4104	545	3,559
Adj R^2	0.603	0.827	0.667	0.859	0.799	0.720

The values in parentheses are t-values adjusted for clustering at the city level. "*", "**" and *** denote significance at the 10%, 5%, and 1% levels, respectively.

demands for technology, talent, and capital compared to the service sector, resulting in higher market entry costs (Forsman 2011). The goal of fostering manufacturing entrepreneurship by relying solely on OGD may be difficult to achieve. Thus, we argue that the positive influence of OGD on entrepreneurship is more evident in the service sector.

To test this, we classified newly registered enterprises into manufacturing and service industries to explore whether the effect of OGD on entrepreneurship differs by industry. We used the industry type as the classification criterion in separate regressions with Model (1). Columns (3)-(4) of Table 10 indicate that *OGD* are only positive and significantly correlated with *Entrep_Serv*. This implies that OGD more effectively promotes service industry entrepreneurship than manufacturing industry entrepreneurship, indicating that potential entrepreneurs are more likely to become service industry entrepreneurs.

Heterogeneity of urban administrative levels. High administrative level cities usually have superior infrastructure, talent, and technological capabilities, along with more developed institutional environments, making them more attractive for new businesses. As a result, high administrative level cities already have more entrepreneurial activities (Tavassoli et al. 2021). Therefore, following the principle of diminishing marginal returns, the effect of OGD is anticipated to be more of "timely assistance" for low administrative level cities rather than "icing on the cake" for high administrative level cities.

We categorized cities into high administrative (including municipalities, provincial capitals, independently planned cities, and special economic zones) and low administrative cities and conducted regression analyses using Model (1). Columns (5)-(6) of Table 10 indicate that for low administrative cities, *OGD* are significantly positively correlated with *Entrep*. However, for high administrative cities, there is no significant correlation between *OGD* and *Entrep*. This implies that low administrative level cities have greater potential for improvement in their entrepreneurial environment compared with high administrative level cities.

Policy synergy effects. The above discussion indicates that OGD improve the entrepreneurial environment and promote entrepreneurship. The innovative city pilot policy aims to boost urban innovation capabilities and enhance the conversion of scientific and technological achievements. As entrepreneurship is derived from innovation and innovation drives entrepreneurship, the two are mutually reinforcing (González-Pernía et al. 2015). Thus, there may be a synergistic effect between OGD and the innovative city pilot policy in facilitating entrepreneurship.

To test this conjecture, we assessed whether the innovative city pilot policy has an entrepreneurial incentive effect by including an interaction term (*OGD* × *Inno*) in Model (1) to test the combined effect of OGD and innovative city pilot policy on regional entrepreneurial activities. *Inno* is a dummy variable that equals 1 for the year a city is selected for the innovative city pilot program and subsequent years and 0 otherwise. Table 11 presents the regression results. The findings reveal that *Inno* is significantly positively correlated with *Entrep*, implying that the innovative city pilot policy stimulates emerging entrepreneurial activities. Moreover, *OGD* × *Inno* is also significantly positively correlated

Table 11 Regression results of synergistic effect.

	(1)	(2)
Variable	***Entrep***	***Entrep***
OGD		0.083
		(1.42)
Inno	0.253**	0.092
	(2.25)	(0.82)
OGD×Inno		0.219**
		(2.15)
Controls	Yes	Yes
Constant	−7.842	−5.139
	(−0.38)	(−0.24)
City FE	Yes	Yes
Year FE	Yes	Yes
N	4104	4104
Adj R^2	0.762	0.763

The values in parentheses are t-values adjusted for clustering at the city level. "**" denotes significance at the 5% level.

with *Entrep*, indicating that combining OGD and innovative city pilot can enhance entrepreneurial activities and create a significant synergistic effect.

Discussion

Conclusions and implications. This study uses the staggered launch of OGD platforms as a natural experiment and takes city-level data from 2009 to 2023 as a sample to conduct empirical analysis to examine the effects of OGD on regional entrepreneurship. Additionally, it analyzes the mechanisms, heterogeneity, and policy synergy effects of this impact. The main conclusions are as follows: (1) OGD enhance regional entrepreneurial activities by improving the entrepreneurial environment, which includes entrepreneurial intention, opportunity, and capital. (2) The effect of OGD on regional entrepreneurship varies significantly across entrepreneurial motivations, industries, and urban administrative levels. Specifically, its impact is more pronounced for opportunity-driven entrepreneurship, the service industry, and low administrative level cities. (3) There is a significant policy synergy between OGD and innovative city pilot initiatives, creating a "$1 + 1 > 2$" effect that stimulates increased entrepreneurial activities.

Based on the above research conclusions, the following policy recommendations are proposed: First, currently, developed countries have a high degree of government data open, while developing countries are still in the early stages. This study shows that OGD effectively stimulate regional entrepreneurship, it has important implications for developing countries, which should seize the opportunities presented by the big data era, actively deploy data open initiatives, and release the value of data elements. Specifically, local governments should actively solicit data needs from societal stakeholders, establish unified metadata standards and open data catalogs, and then enhance the timeliness and accessibility of public data. Local big data bureaus should explore public data licensing and operational models with third parties, ensuring data security while promoting broader, deeper, and higher-quality data sharing. Additionally, improving data literacy and digital skills among local citizens and businesses is crucial. Organizing data application innovation competitions on OGD platforms can encourage public participation, boost entrepreneurial enthusiasm, and increase the likelihood of entrepreneurial success.

Second, mechanism tests reveal that the entrepreneurial environment is an influential channel through which OGD impacts entrepreneurial activities. Therefore, the government should introduce a range of initiatives to create a favorable entrepreneurial environment. This involves creating dedicated zones for enterprise data services and fostering a collaborative relationship between an efficient market and conditionally active government. The government should also expand data openness in areas such as market regulation, enterprise services, and social resources to lower institutional transaction costs, boost administrative efficiency, and reduce entrepreneurs' uncertainty.

Third, the effects of OGD on entrepreneurship vary across different situations. Therefore, the government and relevant departments should consider these differences in policy impacts. Future initiatives for public data openness should focus on improving access to data related to manufacturing, such as information on local industrial enterprises' procurement, production, logistics, and import–export activities. This will help maximize the benefits of open data for manufacturing entrepreneurship. Additionally, efforts should be made to enhance data utilization skills among market participants and lower barriers to accessing public data. Moreover, low administrative level cities should capitalize on OGD by expanding "public data +" applications and speeding up digital government initiatives. This approach will foster widespread entrepreneurship and stimulate local tax revenue, employment, and economic growth.

Theoretical contribution. This study makes several theoretical contributions. First, the relatively large theoretical literature has mostly focused on the traditional factors on entrepreneurial activities, such as formal institutions and informal institutions. This study enhances the current body of research results on factors influencing region entrepreneurship from a data element sharing perspective, which helps expand the academic understanding of entrepreneurship-driving factors. In the digital era, promoting data equality and understanding its impact on entrepreneurs have become critical topics in the theoretical field. However, the current academic attention on these aspects is relatively little. This study examines the impact of data democratization on entrepreneurial activities from the perspective of data element openness and sharing, thus adding to the literature on economic value of data element. Furthermore, it expands the frontiers of transaction cost theory, opportunity theory, and resource dependence theory in the digital era: ① Big data significantly reduces costs in traditional transactional processes—information searching, negotiation, and monitoring; ② The high granularity and real-time capacity of data transform opportunity identification from passive observation to proactive mining; ③ Access to government data fundamentally reshapes resource acquisition pathways and logic under resource dependence theory, substantially enhancing startups' capabilities to obtain external resources.

Limitations and future directions. This study explored the impact of OGD on regional entrepreneurship, utilizing a sample of China's prefecture-level cities. However, it is subject to certain limitations. The current approach of using the launch of government data open platforms as an exogenous shock across regions fails to account for inter-regional variations. Future research could adopt quantitative assessment methods to evaluate regional data openness quality as a measure of local data element sharing levels. This can more accurately capture the situation of OGD. Furthermore, while this paper employs large-sample empirical research methods that reveal generalizable patterns, this approach lacks intuitive depth. Future research could complement this approach by selecting representative cases for in-depth analysis to capture granular insights. Additionally, this

study is based on the transactional cost theory, opportunity theory and resource dependence theory, investigating the mediating role of entrepreneurial intention, opportunity and capital. This only provides some mechanisms for how the OGD influences regional entrepreneurship. Future research can explore more pathways between the two to further enrich the existing research findings. Finally, based on China's government data openness events and the sample of Chinese cities, to determine the applicability of our conclusions to other developing countries, further verification is required.

Data availability

The datasets generated during the current study are available from the corresponding author on reasonable request.

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Author contributions

Haining Zhu: Validation, Visualization, Writing-review & editing. Kaolei Guan: Conceptualization, Supervision, Funding acquisition. Mengting Fu: Data collection and analysis, Methodology, Writing-original draft.

Competing interests

The authors declare no competing interests.

Ethical approval

This article does not contain any studies with human participants performed by any authors.

Informed consent

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Additional information

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